More on the effect of common features

Effect of common features on coincidences

Let's call f a feature that is common to s_1 and to s_2. Suppose that f is not unexpected in itself:

$$C_w(f(s_i)) = C_d(f).$$

Suppose also that f is independent from all other features that characterize s_1 and s_2. This means that the following equality holds:

$$C_w(s_i) = C_w(f(s_i)) + C_w(s_i \mid f(s_i)).$$

If we apply the chain rule on the description side:

$$C_d(s_1 \,\&\, s_2) \leq C_d(f) + C_d(s_1 \mid f) + C_d(s_2 \mid f \,\&\, s_1)$$

We can write (after simplification):

$$U(s_1 \,\&\, s_2) \geq C_w(s_1 \mid f(s_1)) + C_w(f(s_2)) + C_w(s_2 \mid f(s_2)) - C_d(s_1 \mid f) - C_d(s_2 \mid f \,\&\, s_1).$$

The existence of a common feature f introduces a positive term $C_w(f(s_2))$, equal to $C_d(f)$, in the expression of unexpectedness. This explains why common features contribute to coincidences, and why more complex common features are more interesting.

Close